COEP TECHNOLOGICAL UNIVERSITY, PUNE
A Unitary Public University of Government of Maharashtra
(Formerly College of Engineering Pune)
School of Transdisciplinary Sciences & Management
Wellesley Road, Chhatrapati Shivajinagar, Pune - 411005.

Minor Course Title: Industrial & Applied Psychology for Engineers (IAPE)

Overview

Structure	3-4-4-3	**Semesters**	IV, V, VI, VII
Teaching Plan (L-T-P-S) = Semester IV & VII = Semester V & VI =	 3-0-0-1 = 3 4-0-0-1= 4	**MSE** (Continuous Evaluation)	Assignments/Group tasks/Tests/Quizzes- 30 marks Class participation/Teacher evaluation- 20 marks
Credits Semester IV & VII = Semester V & VI =	 3 4	**ESE**	Written exam/Viva-voce/Project work - 50 marks
Upper limit for students = 60 (continues throughout the 4 semesters)			

Course Summary:

This four-semester course is designed to equip engineering students with a comprehensive understanding of psychological principles and their practical applications in industrial and organizational contexts. Blending theory with hands-on practice, the course explores how human behavior, cognition, and mental processes influence workplace dynamics, decision-making, performance, and well-being. Across the semesters, students engage with topics such as individual and group behavior, leadership, team processes, organizational culture, cognitive biases in decision-making, conflict management, and performance prediction through psychometric assessments. The course also integrates emerging areas like behavioral economics and consumer psychology, and addresses vital concerns in occupational health, including stress management, diversity, and mental health.

A strong emphasis is placed on developing applied research skills, including training in research methods, statistical analysis, project work, and ethical research practices. Assessments are diverse—comprising written exams, oral presentations, and project-based evaluations—ensuring students build critical thinking, collaboration, and communication skills. Ultimately, the course prepares students to become psychologically informed engineers who can effectively manage human factors in technical environments, lead diverse teams, and contribute meaningfully to organizational development and innovation.

Semester V

Title: Behaviour at Workplace and Performance Management (BWPM)

Credits: 4

Credits	4	**Semesters**	V
Teaching Plan (L-T-P-S) =	4-0-0-1= 4	**MSE** (Continuous Evaluation)	Assignments/Group tasks/Tests/Quizzes- 30 marks Class participation/Teacher evaluation- 20 marks
		ESE	Written exam/Viva-voce/Project work - 50 marks

Course Outcomes:

By the end of this semester, students will be able to:

1. Analyze the factors that influence team dynamics and leadership and learn strategies for managing conflicts and change within teams.
2. Improve decision-making skills by understanding the cognitive biases in technical environments, while critically evaluating organizational culture and managing change effectively in dynamic work settings.
3. Understand and evaluate techniques of performance assessment and recommend evidence-based training interventions to improve organizational outcomes.
4. Identify key psychological factors affecting occupational health and propose strategies to promote a healthy and inclusive workplace.

Unit	Contents	Lectures (hrs.)
01.	**People at Work** • Group dynamics, Factors affecting group performance; Individual Vs. Group Performance • Teams • Leadership- Leader interaction & situation; Specific leader skills • Group Conflict & Conflict management in technical teams	10
02.	**Decision Making & problem Solving** • Cognitive bias- types, identifying bias, case studies; Techniques to reduce cognitive bias & case studies • Decision making models- types; Decision making in teams- tools & matrices (SWOT, Decision making tree) • Organizational Climate & Culture • Understanding change management • Resistance to change, empowerment & management strategies	12
03.	**Performance Prediction and Management** • Types & Techniques- objective and subjective • Performance evaluation- rating system and process • Psychometric assessment and prediction • Organizational training- Methods • Training evaluation- selection, delivery, transfer of training	12
04.	**Occupational Health** • Workplace stress and coping mechanisms • Work-life balance: Impact of productivity • Fairness, diversity & mental health in workplace	10

Overall Suggested learning resources/references for all semester (continuous updation):

1. Aamodt, M.G. (2013). *Industrial Psychology*. Cengage Learning: Delhi.
2. Wickens, C. D.; Lee, J. D., Liu, Y. & Gordon Becker, S. E. (2015). *An Introduction to Human Factors Engineering*. 2^{nd} Edition. Pearson Education: New Delhi.
3. Landy, F. J. & Conte, J. M. (2010). *Work in the 21^{st} Century: An Introduction to Industrial and Organizational Psychology*. 2^{nd} Edition. Wiley India: New Delhi.
4. Matthewman, L., Rose, A. & Hetherington, A. (2009). *Work Psychology*. Oxford University Press: India.
5. Schultz, D. & Schultz, S. E. (2013). *Psychology and Work Today: An Introduction to Industrial and Organizational Psychology*. 7^{th} Edition. Pearson Education: New Delhi.
6. Schultz, D. & Schultz, S. E. (2002). *Psychology and Work Today*. Pearson Education: New Delhi.

****